

Case Study - Data Scientist - MITCO

Forecasting German DA power prices

1 Objective

Produce the most accurate possible forecast of the hourly day-ahead price of the DE-LU bidding zone (€/MWh) for every hour from 1 September 2026 00:00 to 31 December 2027 23:00.

2 Data provided (folder data/) — the only data allowed

History 1 Jan 2025 → 7 Aug 2026.

File	Content
de_lu_day_ahead_price_hourly.csv	Historical target: DE-LU hourly day-ahead price (mean of the four 15-minute prices since the SDAC switch to 15-minute products on 1 Oct 2025)
de_lu_load_hourly.csv	Actual load and day-ahead load forecast (MW)
de_lu_generation_by_type_hourly.csv	Actual generation per production type (MW) + pumped-storage pumping consumption
de_lu_wind_solar_da_forecast_hourly.csv	Day-ahead forecasts of solar, onshore wind and offshore wind (MW)
de_lu_installed_capacity_yearly.csv	Installed capacity per production type, 2025 and 2026 (MW)
fuel_carbon_prices_daily.csv	THE and TTF day-ahead gas (€/MWh, EEX); EUA front-December future (€/t, EEX). No coal price: to be handled by an explicit assumption

3 Accepted approach

Machine learning: model trained on the fundamentals (load, renewables, fuels/CO2, ...), with an explicit strategy to project the fundamentals over 2026-2027 (climatological profiles, capacity trajectories, fuel scenarios).

4 Constraints

- **No external data** (no forward curves, weather, or other sources). Any 2026-2027 evolution (gas, CO2, capacities, demand, cross-border exchanges...) must be a written and justified assumption,
- Reproducible and optimized code (Python or C++ or both).

5 Deliverables

1. `forecast.csv`: columns `ts_utc`, `ts_local`, `price_eur_mwh`, no missing or duplicated hour, with `p10` and `p90` quantiles,
2. Code + instructions to run it,
3. Analysis note (≤ 2 pages): method, assumptions, backtest, error metrics (hourly MAE and RMSE; MAE of monthly baseload averages, structural realism in terms of negative price hours, seasonality, consistency with fundamentals), limitations, what you would do with more data.